Introduction to Statistics

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What is statistics

- Statistics is the art of learning information from data.

- Statistics is the science that deals with the collection, organization, summarization/presentation, analysis, and interpretation of data to assist in making more effective and reasonable decisions.

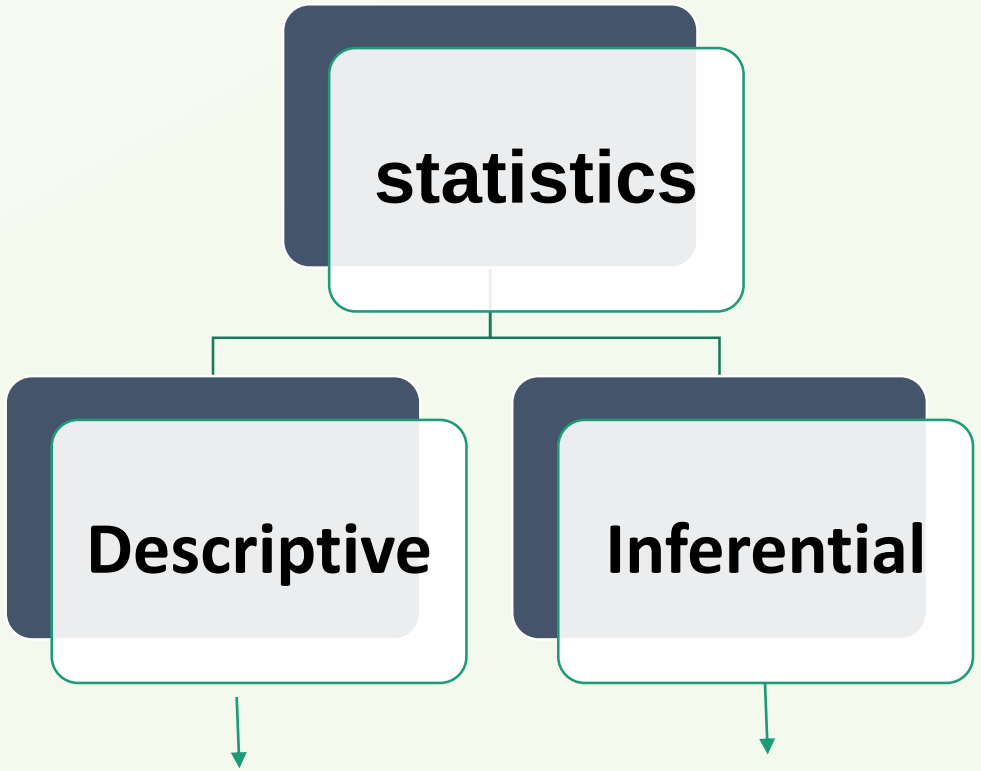
What is statistics

- Data Collection → is a process of gathering information from all the relevant sources
- Data Organization → after data collection, it needs to be organized in a systematic manner
- Data presentation → it's important to present it in a meaningful way
- Data Analysis → involves applying various mathematical and statistical techniques
- Data Interpretation → Once the analysis is done, the results need to be interpreted

Scope of Statistics in life science and engineering

- Machine Learning and Artificial Intelligence
- Data Mining and Big Data
- Software Engineering
- marketing (developing market surveys and strategies for marketing new products)
- public health
- civil engineering

Types of statistics



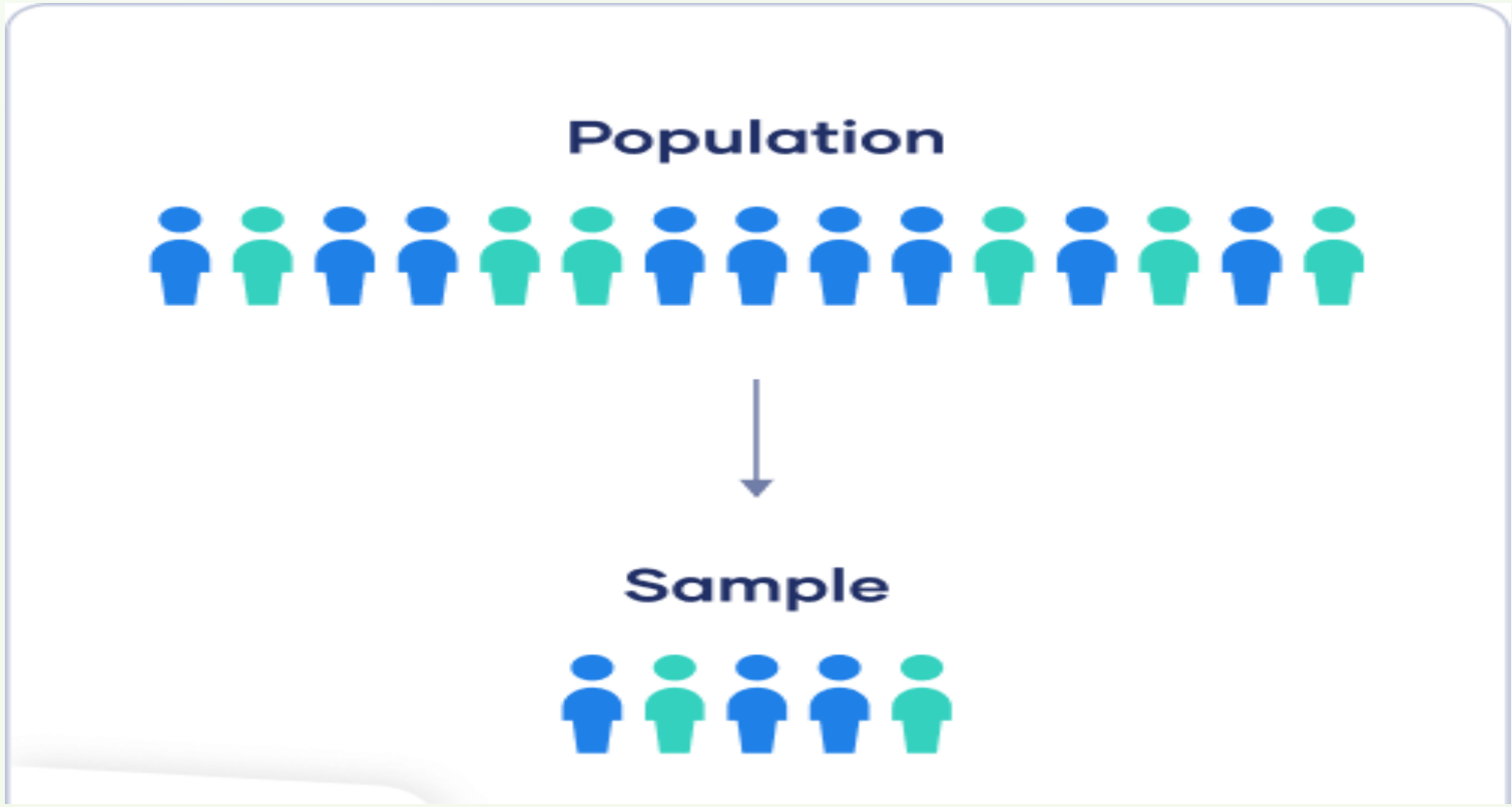
Which is used to summarize, organize, and present a set of data/observations in a meaningful way (e.g., tables, graphs, numerical summaries.

allow to make predictions or inferences about population based on sample.

Population & Sample

- A population is the entire collection of individuals, items, and objects.

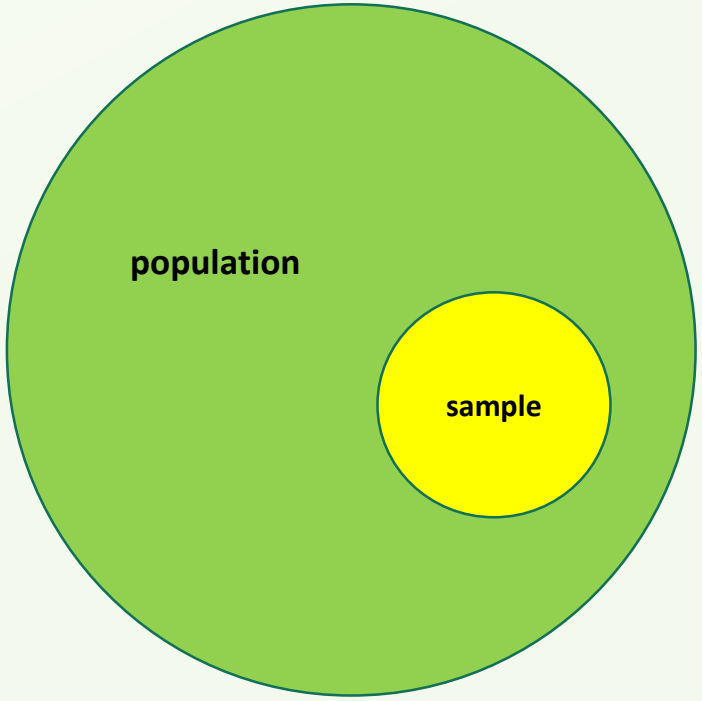
- A small but representative part of the population is called sample.



Census & Survey

- To collect data about an entire population is called “Census”.

- To collect data from the part of the population is called “Survey”.

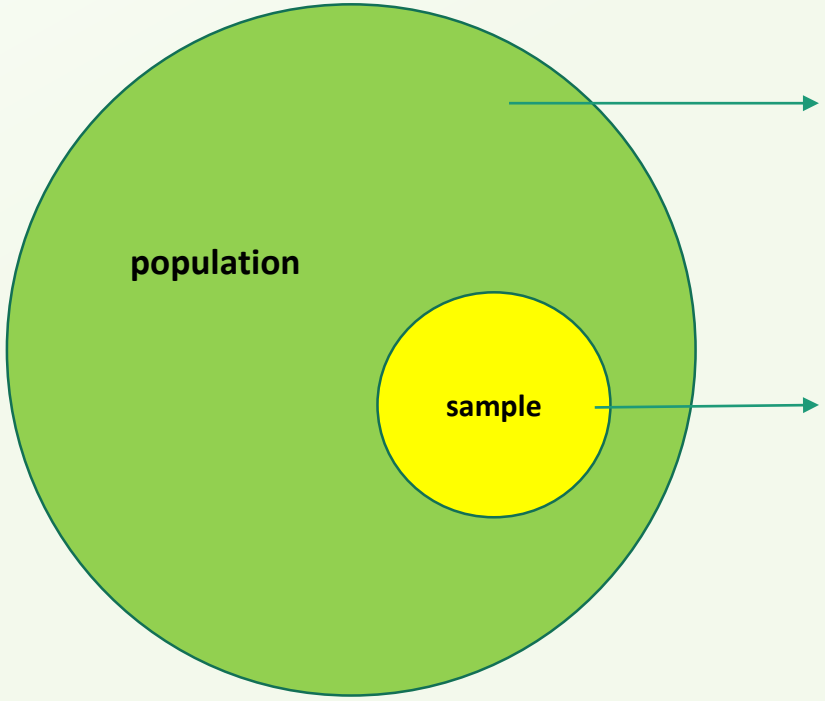


For this example, explain census and survey.

Census & Survey

- To collect data about an entire population is called “Census”.

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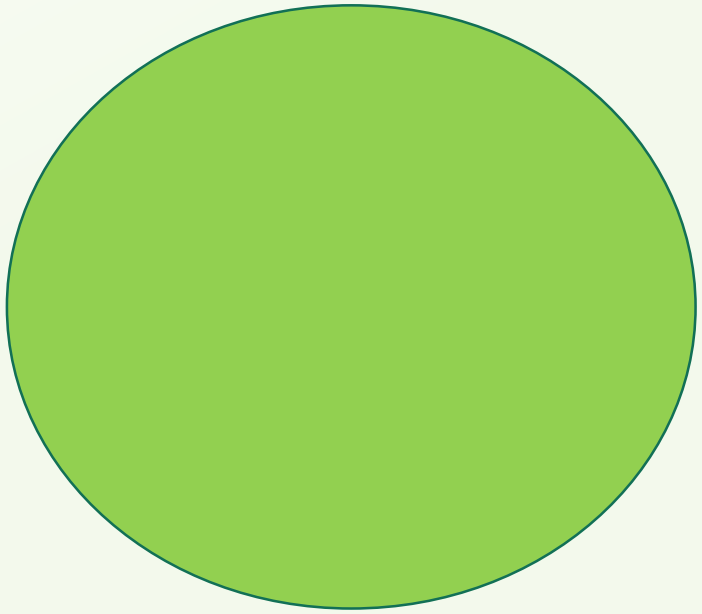
Collects data from this entire circle is “Census”

Collects information from these part is “Survey”

Parameter & Statistic

- **A parameter is a numerical value that describes a characteristic (mean, variance etc.) of a population.**

- **A statistic is a numerical value that describes a characteristic (mean, variance etc.) of a sample.**

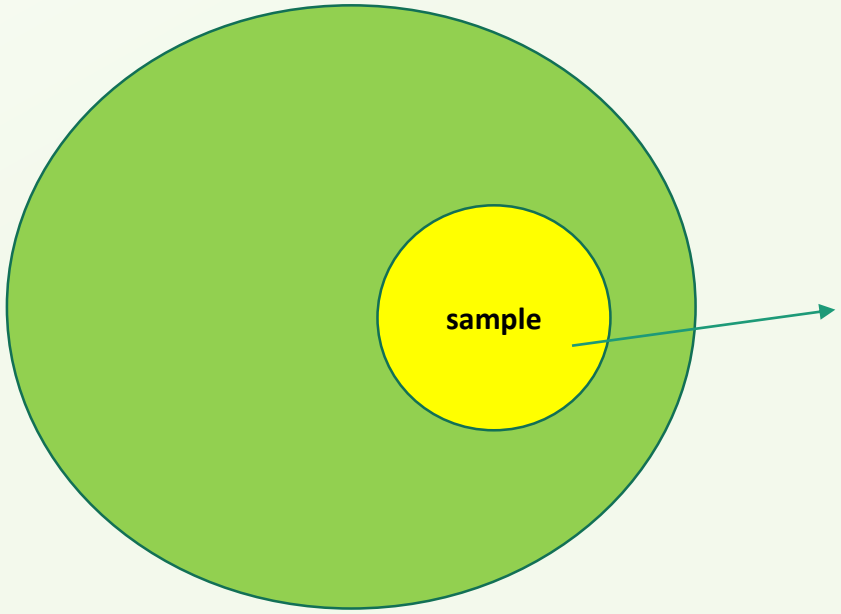


Characteristics of this entire circle is “Parameter”

Parameter & Statistic

- **A parameter is a numerical value that describes a characteristic (mean, variance etc.) of a population.**

- **A statistic is a numerical value that describes a characteristic (mean, variance etc.) of a sample.**



Characteristics of this this circle is “Statistic”

Variable

- If the values of characteristics vary
  - From person to person
  - From object to object
  - From phenomenon to phenomenon
- **Example:** Age, Height, Number of attendance.

Types of Variable

Two types of variable:

1. Qualitative variable (Values can not be measured numerically)

2. Quantitative variable (Values can be measured numerically)

    a. Discrete variable (Countable values)
    b. Continuous variable (Any values within a range)

Example???

Data

- Data refers to information that is collected.
- Collected, measured, and analyzed for various purposes.

Example: Total attendance of each student in this class

13,9,10,8,14

Types of Data

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2. Quantitative variable (Values can be measured numerically)

    a. Discrete variable (Countable values)
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Source of Data

There are two sources of getting statistical data:

1. Primary data (Fresh and First time)
2. Secondary data (Has already been collected by someone)

Scales of Measurements

- Refer to the different ways in which variables or data can be categorized or measured.

- Four measurements
  1. Nominal
  2. Ordinal
  3. Interval
  4. Ratio

Scales of Measurements

Nominal	Ordinal
Must be categorical/qualitative	Must be categorical/qualitative
Can’t be find differences	Can’t be find differences
Can’t be find ratios	Can’t be find ratios
Can’t be ranked	Can be ranked

Scales of Measurements

Interval	Ratio
Must be quantitative	Must be quantitative
Can be ranked	Can be ranked
Zero is not absolute	Zero is absolute
Can be find difference but not ratios	Can be find both difference and ratios

Scales of Measurements

Example	**Property**			Levels
	Order	Difference	Ratio	
Gender				
level of education				
IQ scores				
Person's income				

Scales of Measurements

Example	**Property**			Levels
	Order	Difference	Ratio	
Gender	No	No	No	Nominal
level of education	Yes	No	No	Ordinal
IQ scores	Yes	Yes	No	Interval
Person’s income	Yes	Yes	Yes	Ratio

Scales of Measurements

- **Identify the scale of measurement**
  - Age
  - Temperature in Celsius or Fahrenheit
  - Temperature in Kelvin
  - Socioeconomic status
  - Blood group

ANY QUESTIONS

